



Comparison of machine learning–based algorithms using corneal asymmetry vs. single-metric parameters for keratoconus detection

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Abstract

Purpose To evaluate the diagnostic performance of three different parameter sets relevant to corneal asymmetry in comparison to conventional parameters including maximum anterior corneal curvature ($K_{\max}$) and thinnest corneal thickness for diagnosis of keratoconus.

Methods In this retrospective case control study, 290 eyes with keratoconus and 847 eyes of normal patients were included in the analyses. Corneal tomography data were acquired from Scheimpflug tomography. The sklearn and FastAI libraries were used in a Python 3 environment to create all machine learning models. The original topography metrics and derived metrics together with the clinical diagnoses were used as the dataset for model training. The data were first split to assign 20% of the data to an isolated test set. The remaining data were then split 80/20 to a training and validation group for model training. Sensitivity and specificity outcomes with standard parameters ($K_{\max}$, central curvature, and thinnest pachymetry) and ratio of asymmetry across horizontal, apex centered, and flat axis-centered axis of reflection were studied via various machine learning models.

Results Thinnest corneal pachymetry and $K_{\max}$ were $549.8 \pm 34.3 \mu\text{m}$ and $45.3 \pm 1.7 \text{ D}$ in normal eyes and $460.5 \pm 62.6 \mu\text{m}$ and $59.3 \pm 11.3 \text{ D}$ in keratoconic eyes. Use of only corneal asymmetry ratios across all 4 meridians had mean sensitivity of 99.0% and mean specificity of 94.0%, better than utilizing $K_{\max}$ alone or traditional measures combined ($K_{\max}$, thinnest cornea and inferior-superior asymmetry).

Conclusions By using the ratio of asymmetry between corneal axes alone, a machine learning model could identify patients with keratoconus in our dataset with satisfactory sensitivity and specificity. Further studies on pooled/larger datasets or more borderline population can help validate or refine these parameters.

Keywords Keratoconus · Machine learning · Asymmetry · Detection

Key messages

What is known:

- Keratoconus classification indices frequently use asymmetry to diagnose keratoconus.

What this study adds:

- Use of corneal asymmetry ratios across all 4 meridians had excellent sensitivity and specificity for diagnosis of keratoconus. A machine learning model exhibited accuracy for identifying patients with keratoconus.

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Introduction

Keratoconus is a progressive corneal disorder, marked initially by asymmetric corneal ectasia and thinning [1, 2]. Maximum keratometry readings ($K_{\max}$, defined as

the steepest anterior curvature of the cornea) and thinnest pachymetry values are often used in association with other markers and scores to diagnose keratoconus [3–5]. In addition, the asymmetry of this process of thinning and ectasia has been used as an early marker of keratoconus. Keratoconus classification indices such as the KISA%, the cone-location magnitude index (CLMI) and the Pentacam (Oculus, Wetzlar, Germany)-based indices and the support vector-based BSV index in Sirius (CSO, Italy) use asymmetry to diagnose keratoconus [6–10]. Inferior-superior corneal asymmetry is a frequently used parameter in such classifications. However, considering the inherently rotational near symmetry of the cornea around the apex, it seems prudent that alternative axes are also assessed for asymmetry.

When combining multiple topographical measurements and features to diagnose keratoconus, there are various techniques to convert the numbers into a clinically useful diagnosis. Some are simple scoring algorithms [11], while others use machine learning techniques [12, 13]. The advantage of using machine learning techniques is that it allows complex non-linear modeling between a number of factors, and, rather than programming the modeling manually, the computer is provided with different examples of each diagnosis—allowing it to formulate the optimum algorithm. This approach has been used quite successfully in a number of domains where multiple streams of complex data exist—such as diabetic retinopathy [14], glaucoma [15], and age-related macular degeneration [16]. One disadvantage of machine learning models is that as they increase in complexity, their ability to be explained decreases. When models cannot be explained, this makes them more susceptible to errors when being used on datasets outside of the environment they were created, as it does not allow a clinician an opportunity to review the algorithm and identify clear failings based on domain knowledge. By reducing model complexity, with as few features as necessary for accuracy, models can be built, which are more robust and less sensitive to noise in real-world data.

The aim of this study is to evaluate the diagnostic performance of three different sets of axes based on different reference criteria. We also wanted to assess if the asymmetry in the axis can be compared to other conventional parameters such as maximum keratometry and thinnest corneal thickness.

Methods

This retrospective study was performed at a tertiary care hospital. The study protocol was approved by the Institutional Review Board of the University of Pittsburgh and followed the tenets of the Declaration of Helsinki. Case records were reviewed for clinical data. Corneal tomography data were acquired from Scheimpflug tomography (Pentacam, Oculus,

Wetzlar, Germany). Cases with previous history of ocular surgery including collagen crosslinking, poor-quality scans, corneal scars, severe dry eyes, post keratorefractive surgery ectasia, and pellucid marginal degeneration were excluded. The quality of scans is determined by the Pentacam software. Only scans with good quality (QS: green) were included. All the scans were evaluated by two independent corneal specialists to confirm the diagnosis and the quality. A total of 847 normal eyes and 290 keratoconus eyes were included. In cases with bilateral keratoconus, only one eye was included randomly. In the clinic, we diagnosed keratoconus based on clinical and topographic signs of keratoconus. In addition to the clinical signs and symptoms (including increased astigmatism, slit lamp findings (Fleischer's ring, Munson's or Rizzuti's sign, scissoring, on retinoscopy), the Scheimpflug tomography criteria were used to diagnose keratoconus. Pentacam® has classification criteria based on multiple topographic indices. These criteria are known as TKC (topographic keratoconus classification) [17]. The TKC score for keratoconus ranges from TKC 1–4 with increasing severity. We used the TKC score to classify candidates as normal and keratoconus [17]. Further details of this classification system are elsewhere in literature. "Normal" patients were recruited from contact lens and refractive surgery clinics. As with most machine learning data, increasing the sample size can increase the predictability of the model. Moreover, the decision boundary being evaluated in this study was normal vs. early keratoconus, and therefore we did not subclassify keratoconus as this would have diluted the true positive set and therefore both the training and testing data's robustness. However, for better understanding of our dataset, we have classified it according to the TKC classification as done previously by Herber et al. [17] This data is presented in supplementary Table 1.

Statistical analysis

The tomographic data was exported from the Pentacam using device-generated csv files with the tomographic data (INDEX-LOAD.CSV, SUMMARY-LOAD.CSV, COR-PWR-LOAD.CSV). The data was entered in an Excel worksheet (Microsoft, Richmond, VA) and then transferred to SPSS 16.0 (SPSS Inc., IL) for analysis.

The hemi-corneal axis of reflection was calculated using the following three reference sets (Fig. 1):

- 1) Reference set 1: conventional Cartesian origin: The central cornea was considered the origin (0,0): the four axes derived were conventional 0 to 180°, conventional 90 to 270°, conventional 45 to 225°, and conventional 135 to 315°
- 2) Reference set 2: flat axis centric: The flat central axis was considered horizontal axis. The rationale for this reference is that the flat central axis is the flattest part of

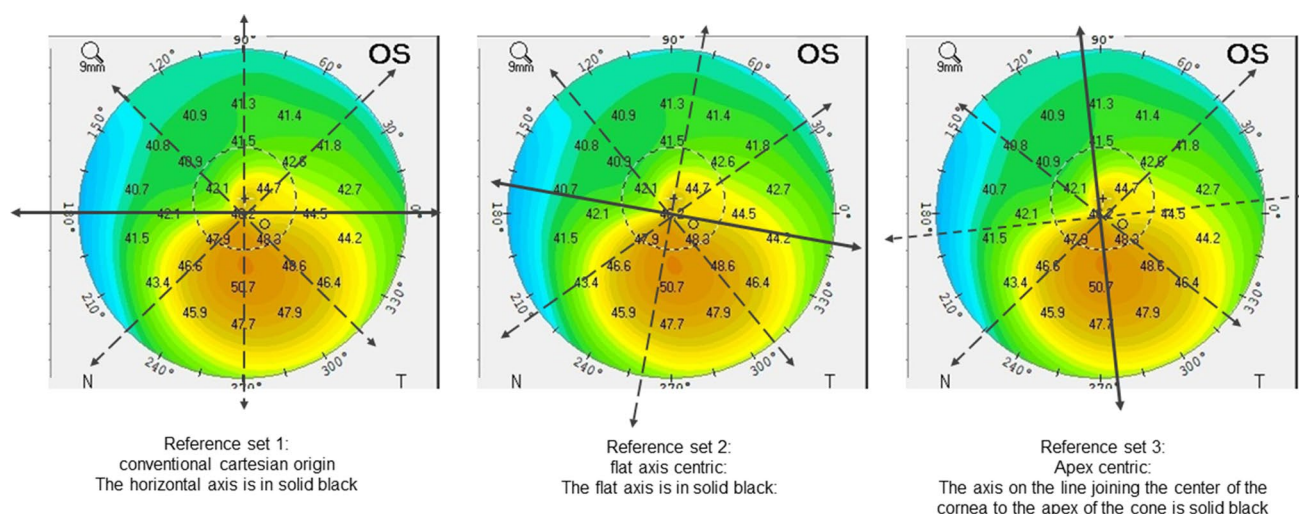


Fig. 1 Schematic diagram explaining three reference sets used to create the hemi-corneal axes of reflection

the cornea and can, therefore, be used as a base reference point. Four axes derived were flat axis referenced 0 to 180°, flat axis referenced 90 to 270°, flat axis referenced 45 to 225°, flat axis referenced 135 to 315°

- 3) Reference set 3: Apex centric: The line joining the center of the cornea and the apex of the cone was considered as the horizontal axis. The rationale for this reference was to customize to each case based on the cone location. Therefore, four axes derived were apex referenced 0 to 180°, apex referenced 90 to 270°, apex referenced 45 to 225°, apex referenced 135 to 315°

Therefore, a total of 12 (3×4) axes were used to create hemi-corneas. The radii of curvature were converted to diopters for each point documented on the csv file. The mean dioptric power for the central 4 mm for each hemicornea was computed. The ratio of the mean dioptric power for each hemicornea was computed.

The above parameters were developed based on the assumption that the cornea gets increasing asymmetry, especially in early to moderate keratoconus and deviates from the normal “milder” asymmetry seen in normal corneas. Specifically for apex centric parameters, the aim was to exploit the asymmetry around the steepest part of the cornea (which is the cone in keratoconus and is, relatively, a more central/nasal point in most normal corneas).

Machine learning

The sklearn and FastAI libraries were used in a Python 3 environment to create all machine learning models. The original topography metrics and derived metrics together with the clinical diagnoses were used as the dataset for model training. The

data were first split to assign 20% of the data to an isolated test set. The remaining data were then split 80/20 to a training and validation group for model training.

We chose to compare a series of models, which are commonly used through research and industry. Simple models such as logistic regression, support vector machines, and K nearest neighbor models were selected for their relative simplicity. Random forest and a gradient boosted decision tree were selected for their superior performance on certain datasets, while remaining simple enough to have explainability. Finally, Neural Net architecture was also explored as this has been found to deal with complex datasets in various medical tasks.

Test data were only seen by the models after model hyper-parameter optimization and not seen by the model otherwise. In order to compare the models, ROC and accuracy metrics were compared. Calculation of p -values was not performed, as we were comparing ROC analysis. The train/validation data split was randomized at all points, further reducing multiple comparison issues.

Table 1 Demographic characteristics in normal and keratoconus patients

	Normal ($n=847$)		Keratoconus ($n=291$)	
	Mean	Standard deviation	Mean	Standard deviation
Age (years)	31.8	8.2	32.5	7.9
Central flat keratometry (D)	43.3	1.5	48.1	7.2
Central steep keratometry (D)	44.5	1.6	52.1	7.8
Central mean keratometry (D)	43.9	1.5	50.0	7.4
Thinnest corneal pachymetry (μm)	549.8	34.3	460.5	62.6
Maximum keratometry (D)	45.3	1.7	59.3	11.3

D, diopters; μm , microns

Table 2 Ratio of reflection across the reference axes in normal and keratoconus patients

Asymmetry assessment reference axes	Normal		Keratoconus	
	Mean	Standard deviation	Mean	Standard deviation
Conventional 0° to 180°	1.011	0.018	1.131	0.088
Conventional 90° to 270°	1.010	0.012	1.054	0.049
Conventional 45° to 225°	1.011	0.010	1.100	0.083
Conventional 135° to 315°	1.010	0.018	1.082	0.066
Apex referenced 0° to 180°	1.011	0.015	1.112	0.079
Apex referenced 90° to 270°	1.010	0.016	1.068	0.067
Apex referenced 45° to 225°	1.010	0.010	1.088	0.075
Apex referenced 135° to 315°	1.010	0.020	1.090	0.079
Flat axis referenced 0° to 180°	1.011	0.016	1.088	0.071
Flat axis referenced 90° to 270°	1.010	0.014	1.096	0.076
Flat axis referenced 45° to 225°	1.010	0.009	1.068	0.070
Flat axis referenced 135° to 315°	1.010	0.009	1.088	0.080

Results

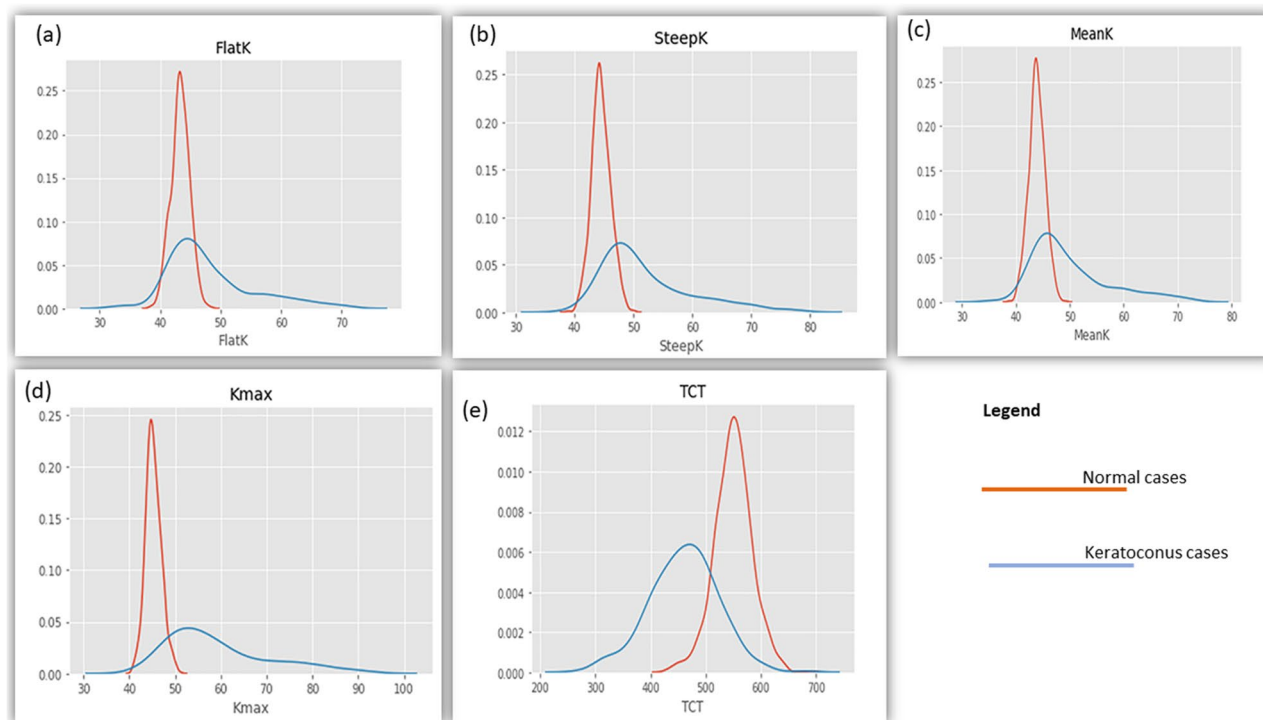
A total of 1137 records were identified and used for the final analysis. This consisted of 290 eyes with keratoconus and 847 eyes of normal patients. In case of bilateral keratoconus, one eye was selected randomly. The demographic

data is given in Table 1. The cross tabulation of our data in terms of severity with the TKC score is in the supplementary Table 1. The standard deviation was significantly higher in the keratoconus group. As shown in Fig. 1, the median distributions of the two data sets (keratoconus and normal) were much closer than what is suggested by the isolated value of the mean only. The ratio of the mean dioptric power for the two hemicorneas for each axis of reflection is given in Table 2.

When we explore the standard keratoconus indices (flat K , flat axis location, mean K , thinnest corneal point, $K_{\max}$, thinnest central thickness, and steep K) in Fig. 2, we note that both the mean and the distribution spread is different between normal and keratoconus patients. When viewing the newly created indices in Fig. 3, we note that this spread in distribution is even wider, and the differences between the two groups become even more appreciable visibly.

When creating the machine learning models, we trialed different sets of inputs to see which inputs are needed to create the optimal model. As seen in Table 3, we found that the use of only asymmetry ratios across all 4 meridians had excellent sensitivity and specificity, better than utilizing $K_{\max}$ alone, or traditional measures ($K_{\max}$, thinnest corneal pachymetry, corneal asymmetry at 180° meridian).

A series of machine learning models were created. A full summary of all models created is seen in Fig. 4, showing

**Fig. 2** Mean and distribution spread between normal and keratoconus patients for the primary parameters

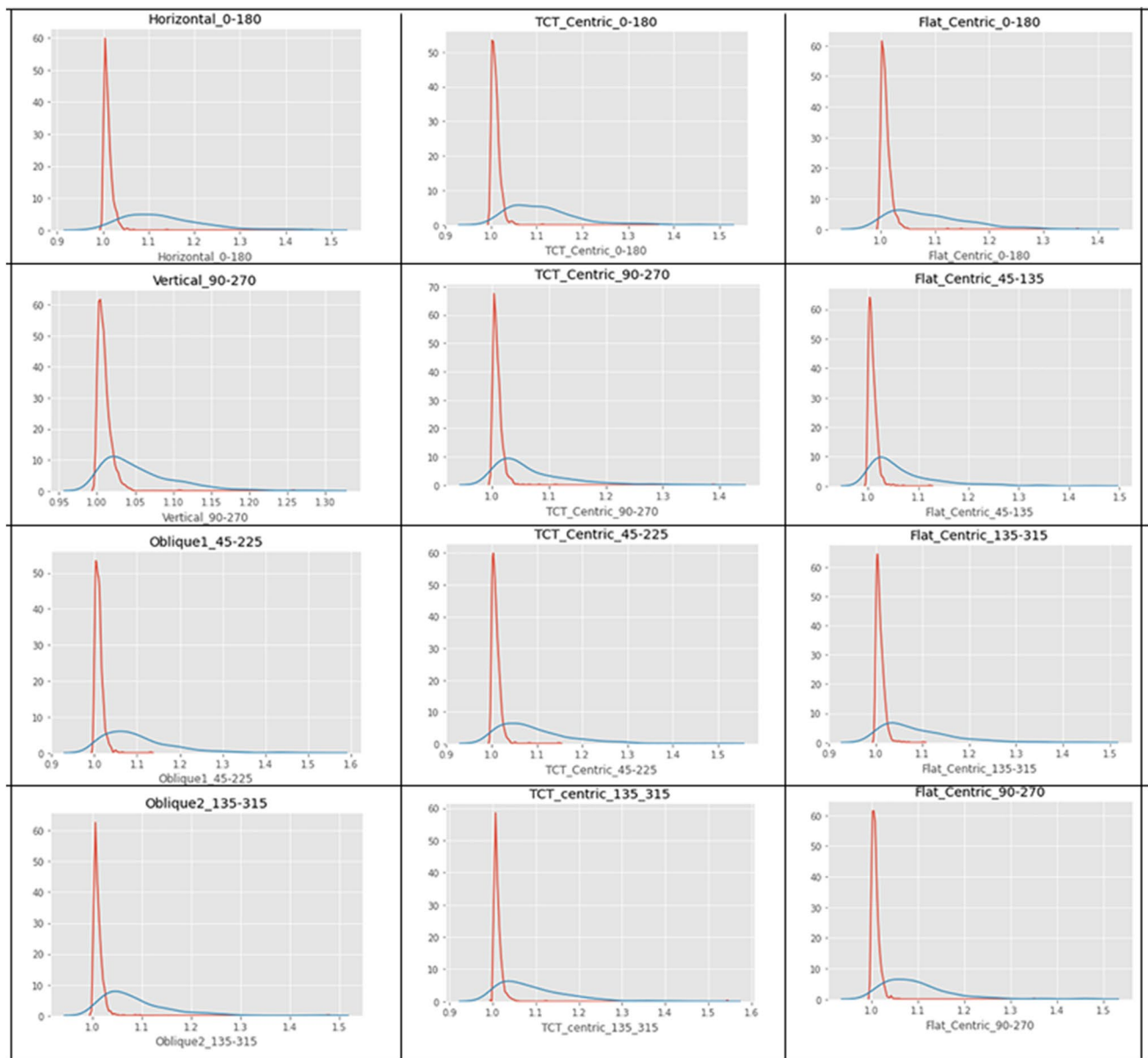


Fig. 3 Mean and distribution spread between normal and keratoconus patients for the asymmetry-based new parameters

relative performance of each model using different feature sets. We found that using corneal asymmetry across all 4 meridians was a simple yet effective feature set for creating our final model. As such, we chose to focus model selection on this feature set. After model training and hyperparameter optimization, we found that random forests provided the best performance balancing sensitivity and specificity, though XGBoost, logistic regression, and neural network architectures also provided reasonable performance. A receiver operator curve for this model is shown in Fig. 5. A feature importance calculation was done. Figure 6 shows the relative weighted importance of each axis in the ultimate model output.

Discussion

Asymmetry of the ectatic cornea has been used to diagnose keratoconus along with maximum value of keratometry and the thinnest pachymetry [7–10]. Our study found that, in a relatively large dataset, the asymmetry of the ectatic cornea has a better discriminant performance compared to apex keratometry and pachymetry.

Most tomography devices have their own classification systems or “hyperparameters” (such as the BAD-D in Pentacam). As hyperparameters can be device specific, we avoided using them in this study to ensure that the data and

Table 3 Average sensitivity/specificity for machine learning models when differentiating by feature set and when differentiating by different model architectures

	Sensitivity	Specificity
Feature sets		
All ratios across 4 meridians	0.990	0.940
Keratometry ratios across 4 meridians	0.991	0.940
All variables	0.981	0.931
Traditional measures	0.989	0.896
Maximum keratometry alone	0.976	0.814
Model architectures:		
Neural network	0.920	0.972
Random forest	0.982	0.928
XGBoost	0.986	0.921
Logistic regression	0.979	0.917
RBF SVM	0.994	0.898
Keratometry nearest neighbors	0.992	0.883
Linear SVM	0.992	0.872

the findings can be a part of a future larger pooled dataset and/or are reproducible in a device-independent fashion.

Even though the inferior-superior asymmetry used conventionally performed very well, the oblique axes were more discriminatory. The performance of the three reference frames was comparable. This is in line with the hypothesis that keratoconus is an asymmetrical, progressive disorder. Use of single-point metrics such as $K_{\max}$ and thinnest corneal thickness could be rather specific criteria if used in the later part of the disease process to classify the severity of

keratoconus. However, early in the course of the disease, where discrimination between a suspect/early case versus a normal eye is needed, these single-point metrics and the secondary parameters derived from them could be less sensitive. The use of asymmetry-based parameters could be a more robust solution as noted in our study.

Other studies have also utilized machine learning to diagnose keratoconus from topography data with varying degrees of success [6, 18, 19]. However, these models are all designed by calculating many metrics (e.g., up to 22 metrics in some) from the topography data and then allowing a machine learning algorithm to incorporate all of those features to predict an outcome. However, when a model with equal accuracy can be created with fewer features, not only is the resultant model more elegant—it tends to be more robust, and generalize better to real-world data. [20] Often, techniques such as principal component analysis are used to simplify more complex models to remove redundant features. Alternatively, as we have done in our model, we have specifically chosen to include only 4 highly discriminant features, which results in excellent model performance, while maintaining explainability and is likely to allow the model to perform better in previously unseen datasets.

We refrained from using unrestricted machine learning (ML) in our analysis. While unrestricted ML algorithms acting on raw data alone do have the ability to make interesting new findings for discrimination, this approach has two major limitations. Firstly, this approach is far more data intensive and requires a much higher number of cases. Secondly, the discriminative factors identified by such ML algorithms cannot be translated into human understandable metrics.

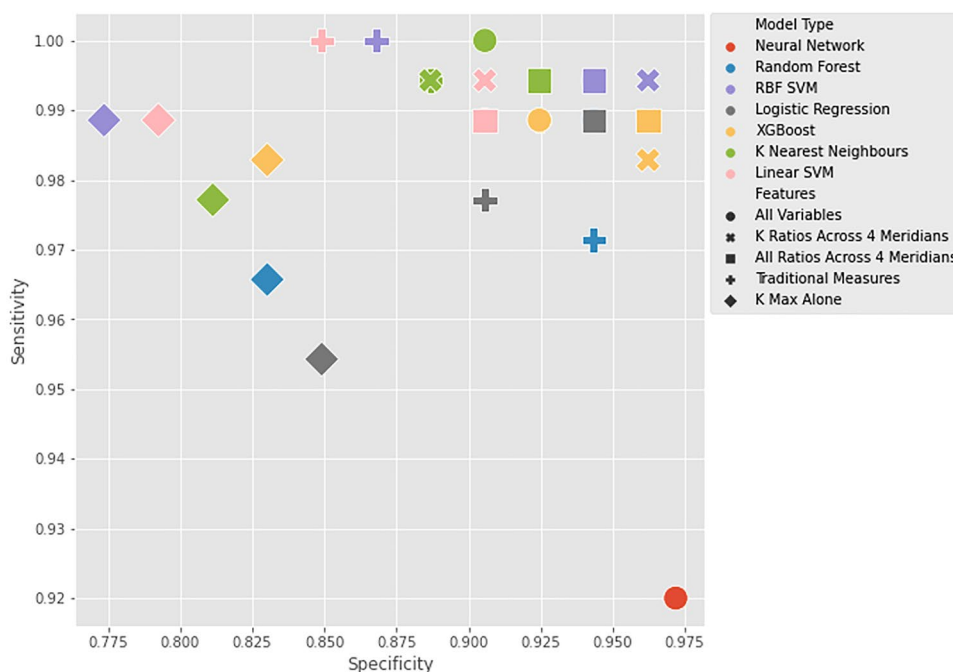
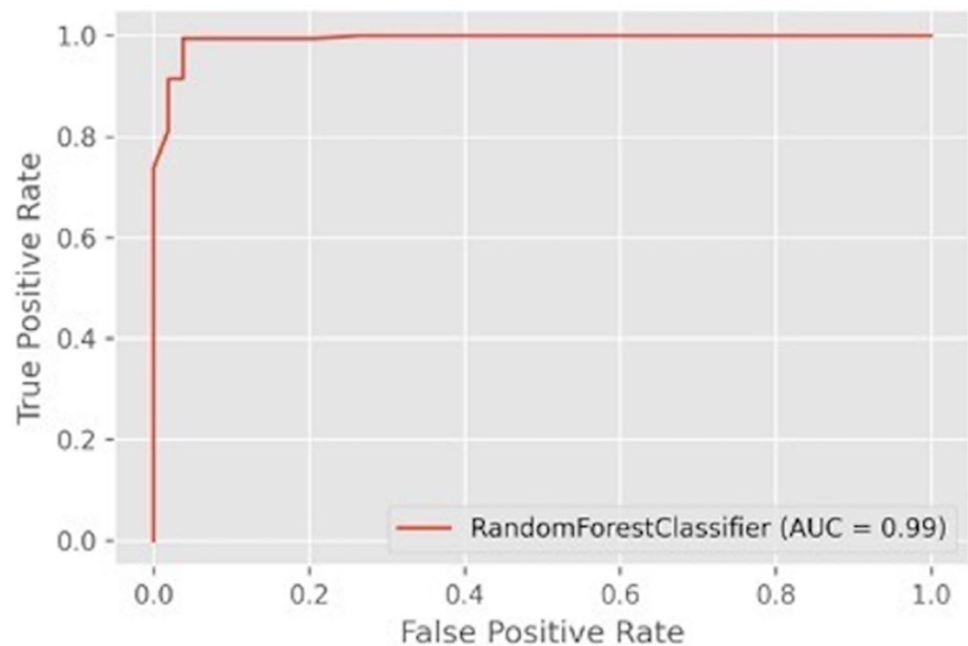
Fig. 4 Sensitivity vs. specificity plots for the machine learning models used

Fig. 5 Receiver operator curve for random forest classifier using only-asymmetry ratios across all four meridians of the cornea



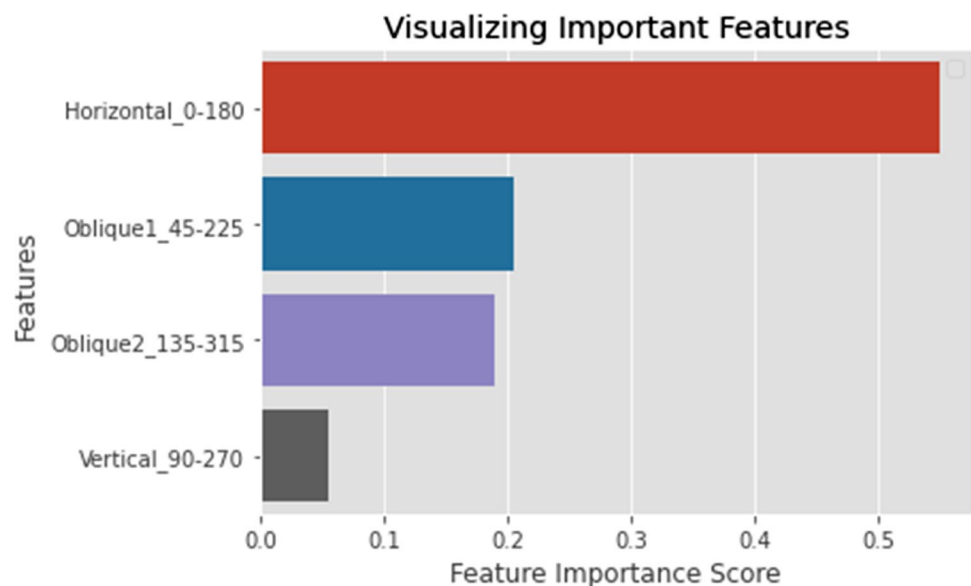
A key tenet of medical AI development is to build models that provide explainable results. By combining numerous factors already understood by clinicians, our model is able to weight these in a complex non-linear manner to provide excellent performance, while requiring minimal input cases and a resulting model, which can be easily explained in a way understandable to clinicians.

Given the relatively simple nature of the models, cross validation was only performed on the winning model, with the other models only using a simple train/validate/test split of data. Five-fold cross validation was performed on the winning model, and we found that there was no change to the model performance compared to the original train/validate/

test split. Theoretically, should there have been variance in the data selected in each test/validate iteration, this would help create a more homogenous result. In the future, with more complex models, cross validation should be used on all models to improve reliability of accuracy metrics.

There are, however, certain shortcomings to our study. As this is a cross-sectional analysis, it will be interesting to see serial progression in keratoconus and the rate of change in asymmetry compared to the rate of change in single-point parameters. As we also noted, increased asymmetry is a relatively sensitive sign, and, therefore, there are chances of false-positive prediction. Localized, mid-peripheral scars could increase the asymmetry and can, potentially, be

Fig. 6 Relative weighted importance of each axis in the ultimate model output



screened as keratoconus suspects. However, slit lamp evaluation in many of these cases can avoid such confounding problems. Similarly, all cases of steep corneas or thin corneas are not keratoconus, and many grading systems give extensive weight to these parameters. Future studies are merited to validate this approach to keratoconus diagnosis in a larger dataset to help further fine-tune model hyperparameters and test its performance across a wider set of pathology. Having said that, initial studies and algorithms need to be on clearly defined data to create better decision boundaries. Our current study is the initial step in this direction and therefore can be labeled as a “proof of concept.” Future studies can look at the performance of this algorithm in more challenging cases.

Finally, having access to a large-scale longitudinal dataset would allow the model to be trained to potentially diagnose keratoconus earlier in the disease process.

In conclusion, we found that using corneal asymmetry across all 4 meridians was a simple yet effective feature set for creating our final model. We found that random forests provided the best performance balancing sensitivity and specificity, though other models (XGBoost, logistic regression, and neural network architectures) also provided reasonable performance.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00417-023-06049-6>.

Declarations

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the (place name of institution and/or national research committee) and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was not needed due to the retrospective study design.

Conflict of interest The authors declare no competing interests.

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